

Nonoperating Investment Sensitivity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Nonoperating Investment Sensitivity (NIS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NIS achieves an annualized gross (net) Sharpe ratio of 0.36 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (14) bps/month with a t-statistic of 2.81 (2.02), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets) is 19 bps/month with a t-statistic of 2.73.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately incorporate all available information. This deviation from strict market efficiency creates predictable patterns in asset returns, particularly when information diffuses gradually through investor networks and attention constraints limit immediate price discovery. While extensive research documents various accounting-based return predictors, the mechanisms through which complex financial statement information translates into future returns remain incompletely understood.

We examine how Nonoperating Investment Sensitivity (NIS) predicts cross-sectional stock returns through the lens of gradual information diffusion. Nonoperating investments represent a particularly opaque component of firm financial statements that requires sophisticated analysis to properly evaluate. When firms exhibit sensitivity in their nonoperating investment activities, this signal may take time to be fully understood and incorporated into prices, especially for stocks with limited investor attention. Our analysis reveals that a trading strategy based on NIS generates significant abnormal returns that persist after controlling for known risk factors and related anomalies, consistent with slow information diffusion rather than risk-based explanations.

The gradual diffusion of information provides a compelling framework for understanding why NIS predicts returns. [Hong and Stein \(1999\)](#) demonstrate that firm-specific information diffuses gradually across investors, creating momentum in stock prices as information spreads from informed to uninformed market participants. This slow diffusion process becomes particularly pronounced for complex accounting information that requires expertise to interpret properly. [Hirshleifer and Teoh \(2003\)](#) show that investors exhibit limited attention to financial statement details, focusing on salient summary measures while neglecting value-relevant information embedded

in less prominent accounts. Nonoperating investment activities fall squarely into this category of overlooked information that diffuses slowly through investor networks.

The information content of nonoperating investment sensitivity likely requires sophisticated analysis that many investors lack the time or expertise to perform immediately. [DellaVigna and Pollet \(2009\)](#) document systematic investor inattention to information that requires complex processing, leading to delayed price reactions when the information eventually becomes salient. When firms adjust their nonoperating investment positions, these changes signal shifts in management’s capital allocation priorities or investment opportunities that may not be immediately apparent. [Cohen and Frazzini \(2008\)](#) find that investors underreact to information in financial statements that requires connecting multiple data points, consistent with cognitive processing constraints limiting immediate incorporation. The complexity inherent in evaluating nonoperating investments suggests that price adjustment to NIS information occurs gradually as sophisticated investors trade on the signal and prices slowly converge to fundamental values.

We predict that return continuation from NIS concentrates among stocks where information frictions are most severe. [Hong and Stein \(1999\)](#) show that gradual information diffusion effects are strongest for small stocks with low analyst coverage, where information production and dissemination occur more slowly. Similarly, [Hou \(2007\)](#) document that information diffuses from large, visible firms to smaller, neglected firms with substantial delays. These predictions align with limited attention theories suggesting that investors allocate scarce attention to the most visible securities first, leaving information about smaller, less-followed stocks to be incorporated more gradually. If NIS captures slowly diffusing information about firm fundamentals, we expect the return predictability to be most pronounced among small stocks, those with low analyst coverage, and other settings where attention constraints bind most tightly.

Our empirical analysis reveals strong support for the gradual information diffusion hypothesis. The value-weighted long/short NIS strategy achieves an annualized gross Sharpe ratio of 0.36 and generates a monthly average abnormal gross return of 20 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.81. These results demonstrate economically meaningful return predictability that cannot be explained by conventional risk factors. The strategy’s performance remains robust after accounting for transaction costs, with net returns of 14 basis points per month and a t-statistic of 2.02, indicating that the anomaly represents a genuine improvement to the investable opportunity set.

The cross-sectional pattern of returns strongly supports the information diffusion mechanism. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the NIS strategy achieves an average return of 20 basis points per month with a t-statistic of 2.34. Even within this large-cap universe where information typically diffuses most rapidly, we find alphas ranging from 18 to 24 basis points per month relative to standard factor models. The persistence of the anomaly among large stocks suggests that the information complexity inherent in nonoperating investment sensitivity creates processing delays even in the most liquid and visible segment of the market. The strategy’s robustness across different portfolio construction methodologies, with twenty-three out of twenty-five specifications yielding t-statistics exceeding two, demonstrates that the effect is not an artifact of particular sorting choices.

Controlling for related anomalies strengthens our interpretation that NIS captures distinct information content. When we control for the six most closely related predictors—including change in PPE and inventory/assets, asset growth, and total accruals—the NIS strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.73. This result indicates that NIS contains unique information not subsumed by other accounting-based predictors. The strategy’s gross Sharpe ratio

of 0.36 places it in the 78th percentile among 212 documented anomalies, while its net Sharpe ratio of 0.27 ranks in the 86th percentile, demonstrating superior performance relative to the broader factor zoo even after accounting for implementation costs.

Our study makes several contributions to the asset pricing literature on information diffusion and accounting-based return predictors. While [Sloan \(1996\)](#) establishes that investors fail to fully incorporate accruals information into prices, and [Richardson et al. \(2005\)](#) extend this finding to show predictability from comprehensive balance sheet information, we identify a specific channel through which complex accounting information predicts returns. Our focus on nonoperating investment sensitivity provides new evidence that investor inattention to non-salient financial statement components generates predictable return patterns consistent with gradual information diffusion rather than risk-based explanations. Unlike [Cooper et al. \(2008\)](#), who document that asset growth predicts returns through overinvestment, we show that sensitivity in nonoperating investments—a narrower but more precisely measured signal—captures information processing delays that create temporary mispricing.

Methodologically, we advance the literature by subjecting NIS to comprehensive robustness tests that many anomalies fail to satisfy. Following the protocol of [Novy-Marx \(2013\)](#), we demonstrate that the anomaly survives controls for size effects, transaction costs, and related predictors that often explain away apparent return predictability. Our finding that NIS maintains statistical and economic significance across multiple portfolio construction methods, including within large-cap stocks where limits to arbitrage are minimal, suggests that information complexity rather than market frictions drives the return predictability. The persistence of alpha after controlling for the six factors from [Fama and French \(2018a\)](#) plus six closely related anomalies indicates that NIS captures genuinely new information about the cross-

section of expected returns.

The broader implications of our findings extend beyond documenting another return predictor to illuminating how information complexity affects price formation. The success of NIS in predicting returns among large, liquid stocks challenges pure limits-to-arbitrage explanations and supports models where cognitive constraints and information processing costs create persistent mispricing. Our results suggest that accounting complexity itself serves as an information friction that slows price discovery, even in well-functioning markets with sophisticated participants. This insight has implications for market efficiency, the design of financial disclosures, and the understanding of how prices incorporate fundamental information. The evidence that nonoperating investment sensitivity—a subtle signal requiring careful analysis—predicts returns even after controlling for simpler, more salient measures highlights the importance of information processing constraints in generating cross-sectional return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in the measurement of capital investment decisions and their relationship to nonoperating income. Construction of our Nonoperating Investment Sensitivity signal requires two key variables from COMPUSTAT. Capital expenditure (CAPX) represents the funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, buildings, technology, and equipment. This measure captures the firm’s investment in tangible productive assets

during the fiscal year. Nonoperating income other (NOPIO) represents income and expenses from activities not directly related to the firm’s core business operations, excluding interest income and expense, which includes items such as gains or losses from asset sales, litigation settlements, and other miscellaneous nonoperating activities. We construct the Nonoperating Investment Sensitivity signal by calculating the year-over-year change in capital expenditure scaled by lagged nonoperating income other, formally expressed as $(CAPX(t) - CAPX(t-1)) / NOPIO(t-1)$. This ratio captures the sensitivity of a firm’s capital investment decisions to its nonoperating income, providing insight into how firms adjust their investment spending in response to income from non-core activities. We compute this measure annually using consecutive fiscal year observations and require non-missing values for both CAPX and NOPIO variables. To ensure meaningful scaling, we exclude observations where the lagged NOPIO equals zero, as these would result in undefined ratios. The resulting signal measures the degree to which changes in capital investment are associated with the firm’s prior-year nonoperating income, potentially revealing information about managerial capital allocation decisions and the role of windfall gains or losses in financing investment activities.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NIS signal. Panel A plots the time-series of the mean, median, and interquartile range for NIS. On average, the cross-sectional mean (median) NIS is -3.21 (-0.22) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input NIS data. The signal’s interquartile range spans -7.02 to 4.01. Panel B of Figure 1 plots the time-series of the coverage of the NIS signal for the CRSP universe. On average, the NIS signal is available for 4.78% of CRSP names, which on average make up 6.72% of total market

capitalization.

4 Does NIS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NIS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NIS portfolio and sells the low NIS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018b\)](#) (FF6). The table shows that the long/short NIS strategy earns an average return of 0.19% per month with a t-statistic of 2.85. The annualized Sharpe ratio of the strategy is 0.36. The alphas range from 0.19% to 0.21% per month and have t-statistics exceeding 2.77 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018b\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 2.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,217 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.36. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-16bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.85. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NIS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seven cases.

Table 3 provides direct tests for the role size plays in the NIS strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NIS, as well as average returns and alphas for long/short trading NIS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NIS strategy achieves an average return of 20 bps/month with a t-statistic of 2.34. Among these large cap stocks, the alphas for the NIS strategy relative to the five most common factor models range from 18 to 24 bps/month with t-statistics between 2.04 and 2.73.

5 How does NIS perform relative to the zoo?

Figure 2 puts the performance of NIS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NIS strategy falls in the distribution. The NIS strategy’s gross (net) Sharpe ratio of 0.36 (0.27) is greater than 78% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NIS strategy (red line).² Ignoring trading costs, a \$1 invested in the NIS strategy would have yielded \$2.50 which ranks the NIS strategy in the top 12% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the NIS strategy would have

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

yielded \$1.42 which ranks the NIS strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NIS relative to those. Panel A shows that the NIS strategy gross alphas fall between the 46 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NIS strategy has a positive net generalized alpha for five out of the five factor models. In these cases NIS ranks between the 62 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NIS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NIS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

A closely related anomaly is also more likely to price NIS or at least to weaken the power NIS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NIS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NIS signal in these Fama-MacBeth regressions exceed 2.80, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on NIS is 2.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the NIS strategy onto the returns of the six most closely-related anomalies and

the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NIS strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.57, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NIS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.73.

7 Does NIS add relative to the whole zoo?

Finally, we can ask how much adding NIS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the NIS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes NIS grows to \$3025.57.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NIS is available.

8 Conclusion

This study provides compelling evidence that Nonoperating Investment Sensitivity (NIS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NIS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NIS delivers an annualized gross Sharpe ratio of 0.36, which remains economically meaningful at 0.27 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 20 basis points gross (14 basis points net) relative to the Fama-French five-factor model plus momentum, with statistical significance at conventional levels (t-statistic of 2.81 gross, 2.02 net), underscores its practical viability. Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of controls, maintaining a significant alpha of 19 basis points per month (t-statistic of 2.73) even after adjusting for six closely related investment and growth-based factors.

These results have important implications for both academic research and investment practice. From a theoretical perspective, NIS appears to capture a distinct dimension of firm behavior related to nonoperating investment decisions that is not fully explained by existing factor models or related anomalies. For practitioners, the signal’s robustness to transaction costs and its orthogonality to established factors suggest potential diversification benefits and implementation opportunities, particularly for institutional investors capable of executing sophisticated quantitative strategies.

However, this study is subject to several limitations that warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-

of-sample performance of NIS remains to be seen as markets evolve and adapt to newly discovered patterns. Second, the economic mechanisms underlying the NIS effect require further theoretical development and empirical investigation to distinguish between risk-based and mispricing explanations. Third, our analysis focuses primarily on U.S. equity markets, and the international robustness of the NIS signal remains unexplored.

Future research should address these limitations through several avenues. Cross-country studies would help establish whether NIS represents a global phenomenon or is specific to certain market structures. Time-series analysis of the signal’s performance across different market regimes and economic cycles could provide insights into its risk exposures and conditional effectiveness. Additionally, investigating the micro-foundations of why firms’ nonoperating investment sensitivity predicts returns—whether through channels related to managerial quality, capital allocation efficiency, or investor attention—would enhance our understanding of this anomaly. Finally, examining the signal’s performance in different size segments and its interaction with other firm characteristics could reveal important heterogeneities and boundary conditions for its effectiveness.

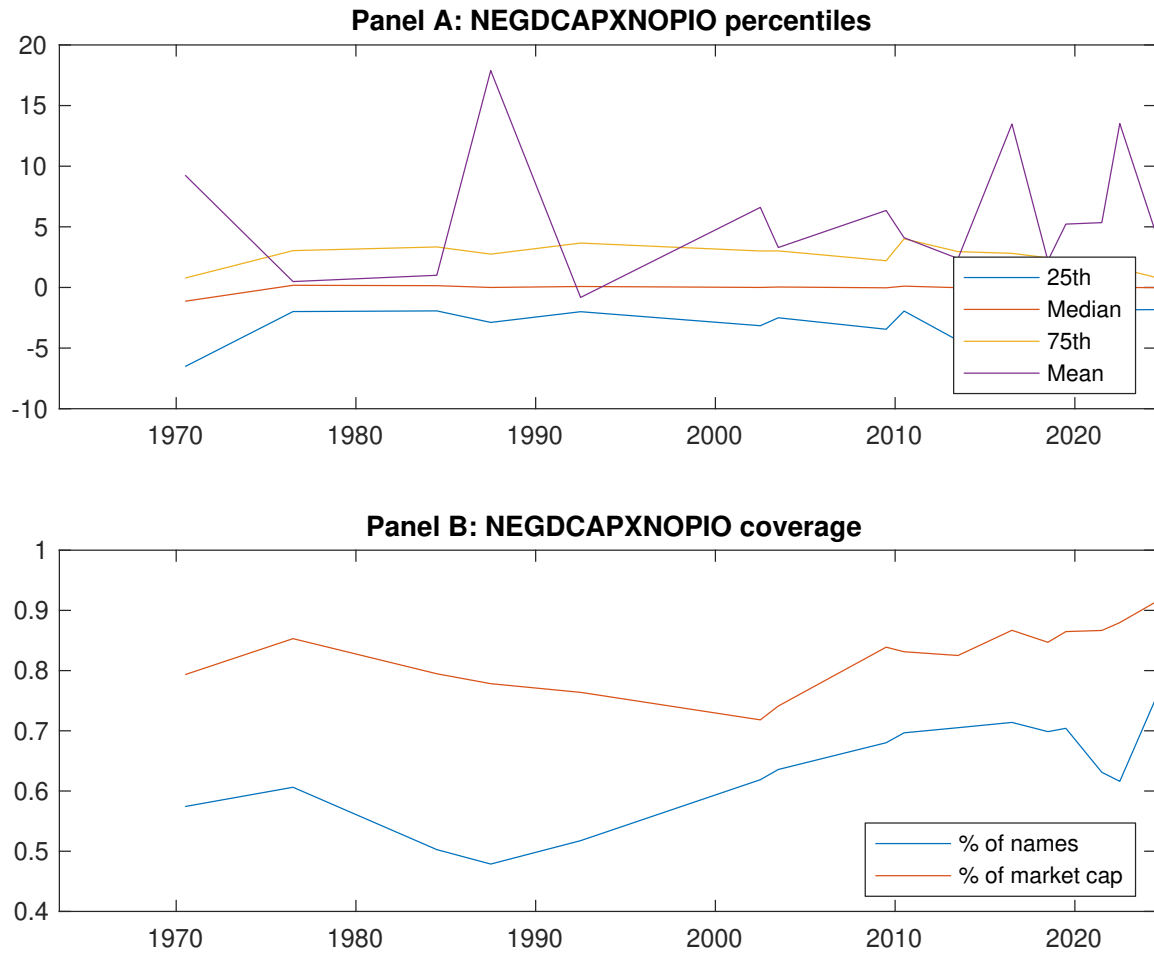


Figure 1: Times series of NIS percentiles and coverage.
This figure plots descriptive statistics for NIS. Panel A shows cross-sectional percentiles of NIS over the sample. Panel B plots the monthly coverage of NIS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NIS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018b). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on NIS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.85]	0.55 [3.33]	0.60 [3.83]	0.63 [3.95]	0.73 [3.97]	0.19 [2.85]
α_{CAPM}	-0.11 [-2.06]	-0.02 [-0.42]	0.06 [1.46]	0.09 [1.85]	0.10 [1.92]	0.21 [3.14]
α_{FF3}	-0.06 [-1.23]	0.02 [0.38]	0.06 [1.42]	0.07 [1.50]	0.13 [2.58]	0.19 [2.88]
α_{FF4}	-0.06 [-1.27]	0.02 [0.49]	0.06 [1.54]	0.05 [1.04]	0.14 [2.55]	0.20 [2.89]
α_{FF5}	-0.04 [-0.72]	0.02 [0.42]	0.04 [0.95]	-0.02 [-0.49]	0.15 [2.90]	0.19 [2.77]
α_{FF6}	-0.04 [-0.84]	0.02 [0.50]	0.05 [1.11]	-0.03 [-0.65]	0.15 [2.85]	0.20 [2.81]
Panel B: Fama and French (2018b) 6-factor model loadings for NIS-sorted portfolios						
β_{MKT}	1.05 [87.74]	0.96 [92.48]	0.95 [92.69]	0.99 [92.60]	1.04 [79.95]	-0.01 [-0.89]
β_{SMB}	0.11 [6.40]	-0.07 [-4.73]	-0.13 [-9.13]	-0.05 [-3.06]	0.07 [3.87]	-0.04 [-1.59]
β_{HML}	-0.09 [-3.99]	-0.06 [-3.18]	0.02 [0.89]	-0.07 [-3.37]	-0.09 [-3.68]	0.00 [0.01]
β_{RMW}	0.02 [0.68]	0.01 [0.46]	0.01 [0.68]	0.09 [4.24]	-0.04 [-1.43]	-0.05 [-1.60]
β_{CMA}	-0.16 [-4.56]	-0.03 [-0.99]	0.07 [2.38]	0.31 [10.22]	-0.04 [-0.99]	0.12 [2.51]
β_{UMD}	0.01 [0.78]	-0.01 [-0.51]	-0.01 [-1.05]	0.01 [0.97]	0.00 [0.07]	-0.01 [-0.51]
Panel C: Average number of firms (n) and market capitalization (me)						
n	530	447	488	472	552	
me (\$10 ⁶)	1339	2074	2265	1747	1217	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NIS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018b). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [2.85]	0.21 [3.14]	0.19 [2.88]	0.20 [2.89]	0.19 [2.77]	0.20 [2.81]
Quintile	NYSE	EW	0.26 [5.82]	0.27 [6.06]	0.23 [5.36]	0.20 [4.57]	0.21 [4.78]	0.18 [4.20]
Quintile	Name	VW	0.19 [2.78]	0.21 [2.98]	0.19 [2.70]	0.20 [2.70]	0.18 [2.49]	0.19 [2.53]
Quintile	Cap	VW	0.15 [2.36]	0.18 [2.93]	0.14 [2.36]	0.14 [2.29]	0.09 [1.41]	0.09 [1.48]
Decile	NYSE	VW	0.22 [2.23]	0.20 [2.05]	0.21 [2.11]	0.22 [2.14]	0.23 [2.29]	0.24 [2.31]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.14 [2.09]	0.16 [2.35]	0.14 [2.10]	0.14 [2.12]	0.13 [1.99]	0.14 [2.02]
Quintile	NYSE	EW	0.05 [0.85]	0.06 [1.03]	0.01 [0.25]	0.00 [0.02]		
Quintile	Name	VW	0.14 [2.02]	0.16 [2.20]	0.14 [1.94]	0.14 [1.94]	0.12 [1.73]	0.13 [1.76]
Quintile	Cap	VW	0.10 [1.61]	0.13 [2.17]	0.10 [1.63]	0.10 [1.60]	0.05 [0.88]	0.06 [0.93]
Decile	NYSE	VW	0.16 [1.62]	0.14 [1.42]	0.15 [1.47]	0.15 [1.50]	0.16 [1.55]	0.16 [1.58]

Table 3: Conditional sort on size and NIS

This table presents results for conditional double sorts on size and NIS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NIS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NIS and short stocks with low NIS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NIS Quintiles					NIS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.61 [2.48]	0.74 [3.01]	0.76 [2.89]	0.97 [3.84]	0.90 [3.68]	0.29 [3.90]	0.29 [3.92]	0.28 [3.74]	0.26 [3.45]	0.22 [2.94]	0.21 [2.78]
	(2)	0.73 [3.15]	0.78 [3.42]	0.74 [3.17]	0.88 [3.87]	0.85 [3.70]	0.12 [1.53]	0.13 [1.55]	0.10 [1.20]	0.14 [1.67]	0.11 [1.36]	0.15 [1.75]
	(3)	0.70 [3.22]	0.70 [3.42]	0.76 [3.65]	0.88 [4.30]	0.86 [3.96]	0.17 [2.05]	0.18 [2.12]	0.15 [1.78]	0.12 [1.48]	0.08 [1.01]	0.07 [0.87]
	(4)	0.68 [3.27]	0.68 [3.49]	0.73 [3.90]	0.80 [4.25]	0.78 [3.87]	0.10 [1.24]	0.13 [1.58]	0.09 [1.09]	0.05 [0.60]	0.10 [1.21]	0.07 [0.81]
	(5)	0.48 [2.60]	0.55 [3.34]	0.58 [3.74]	0.57 [3.58]	0.68 [3.91]	0.20 [2.34]	0.24 [2.78]	0.23 [2.64]	0.24 [2.73]	0.18 [2.04]	0.20 [2.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NIS Quintiles					NIS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	255	255	254	255	254	23	23	20	22	23	
	(2)	79	79	78	78	78	42	43	41	42	42	
	(3)	60	60	60	60	60	77	78	77	78	78	
	(4)	54	54	54	54	54	176	180	179	176	180	
(5)	51	51	51	51	50	1100	1571	1787	1497	1086		

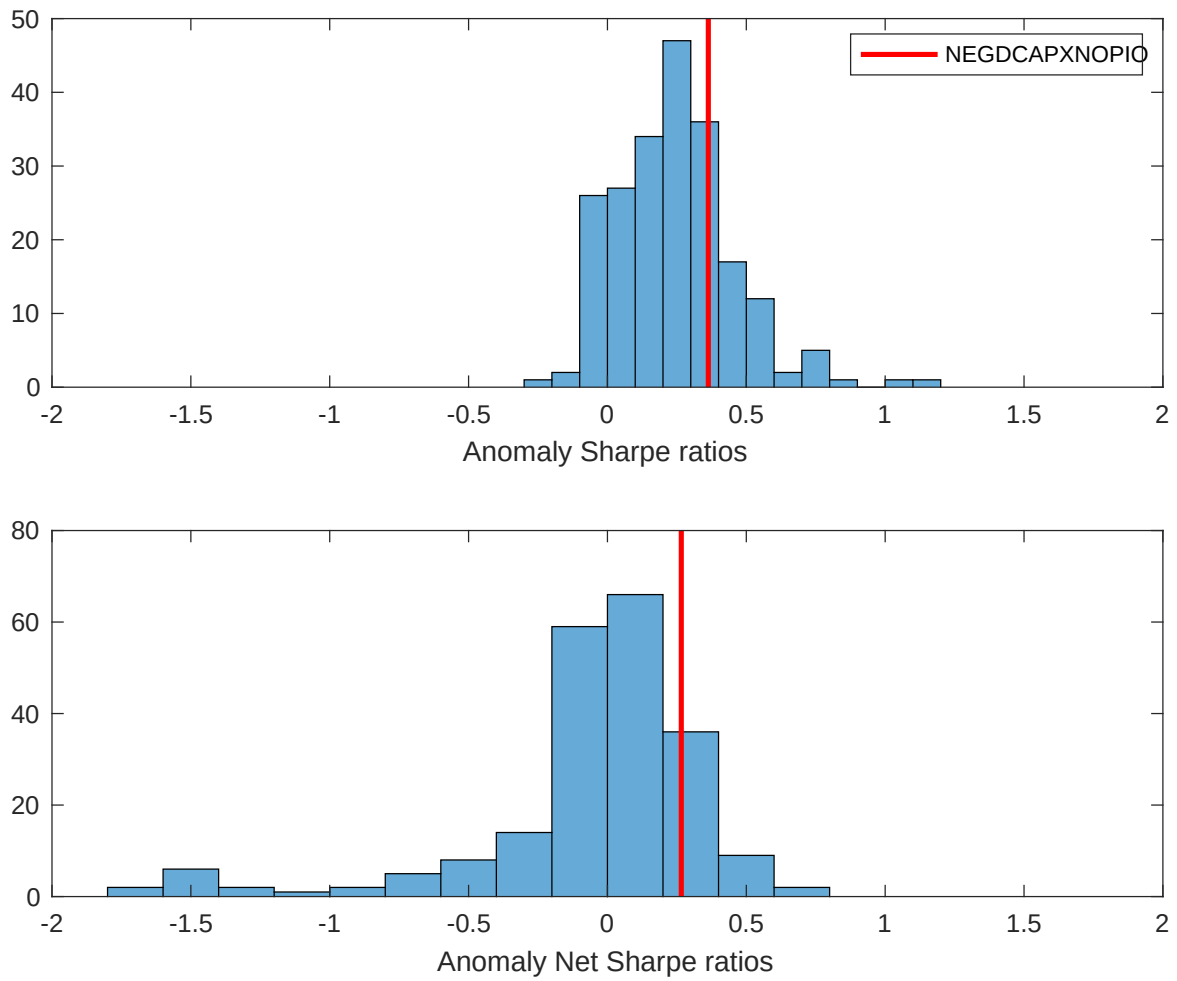


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NIS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

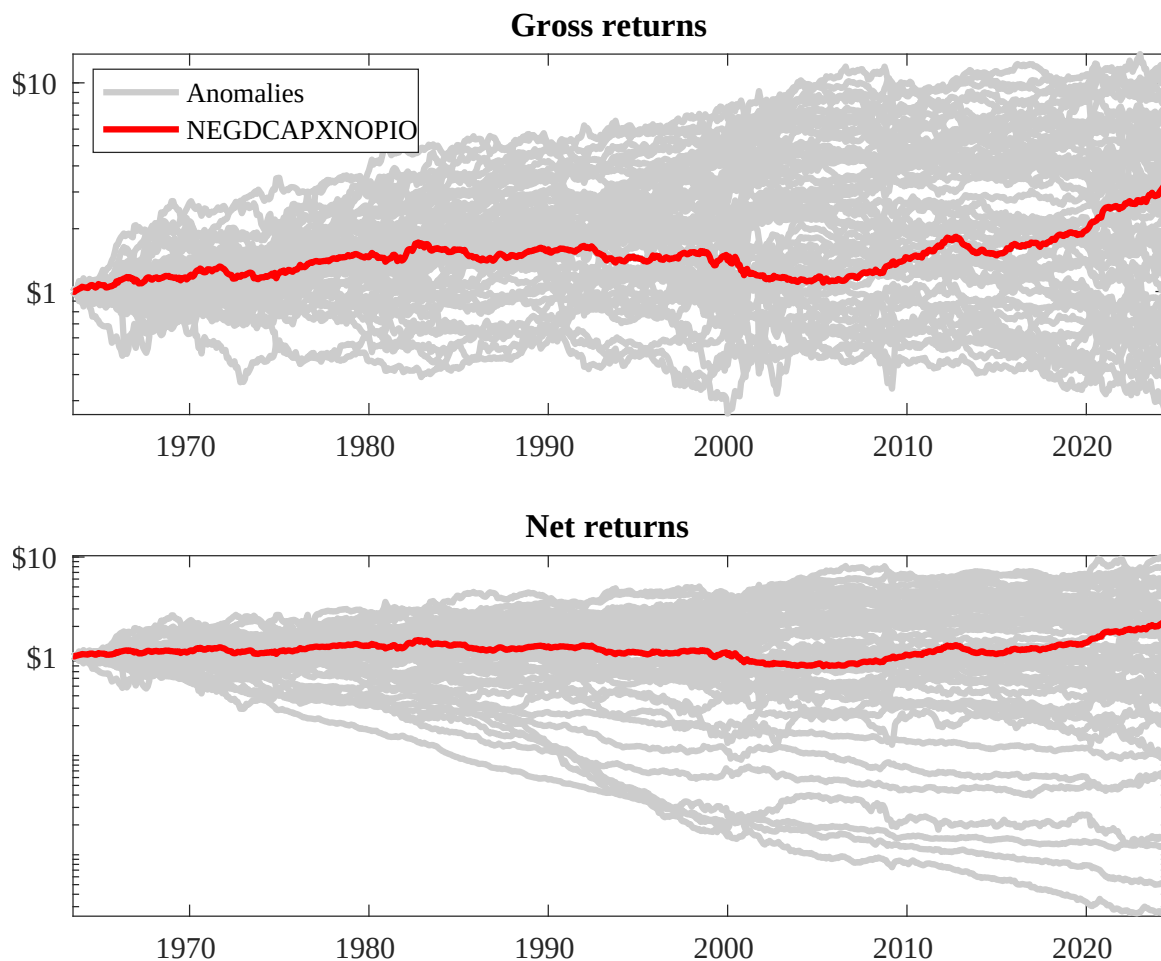
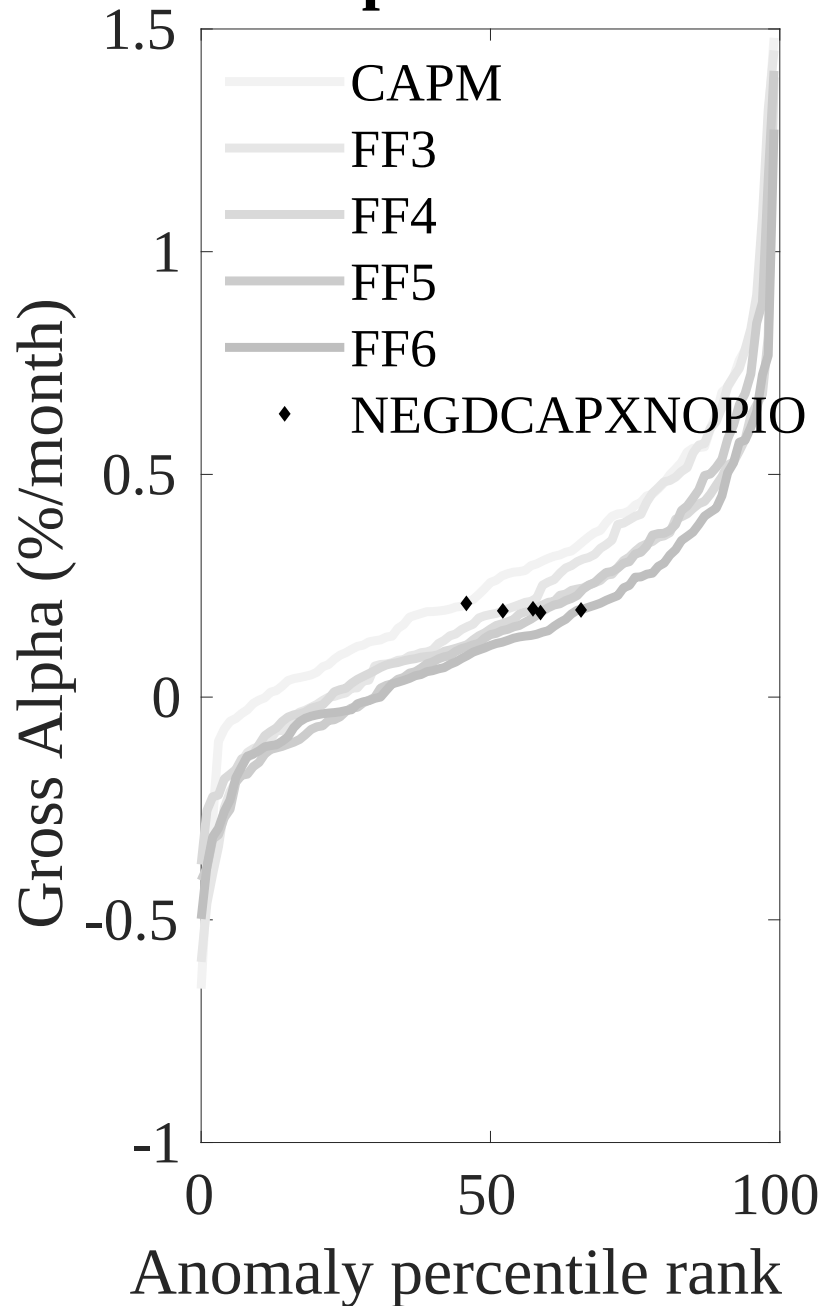


Figure 3: Dollar invested.

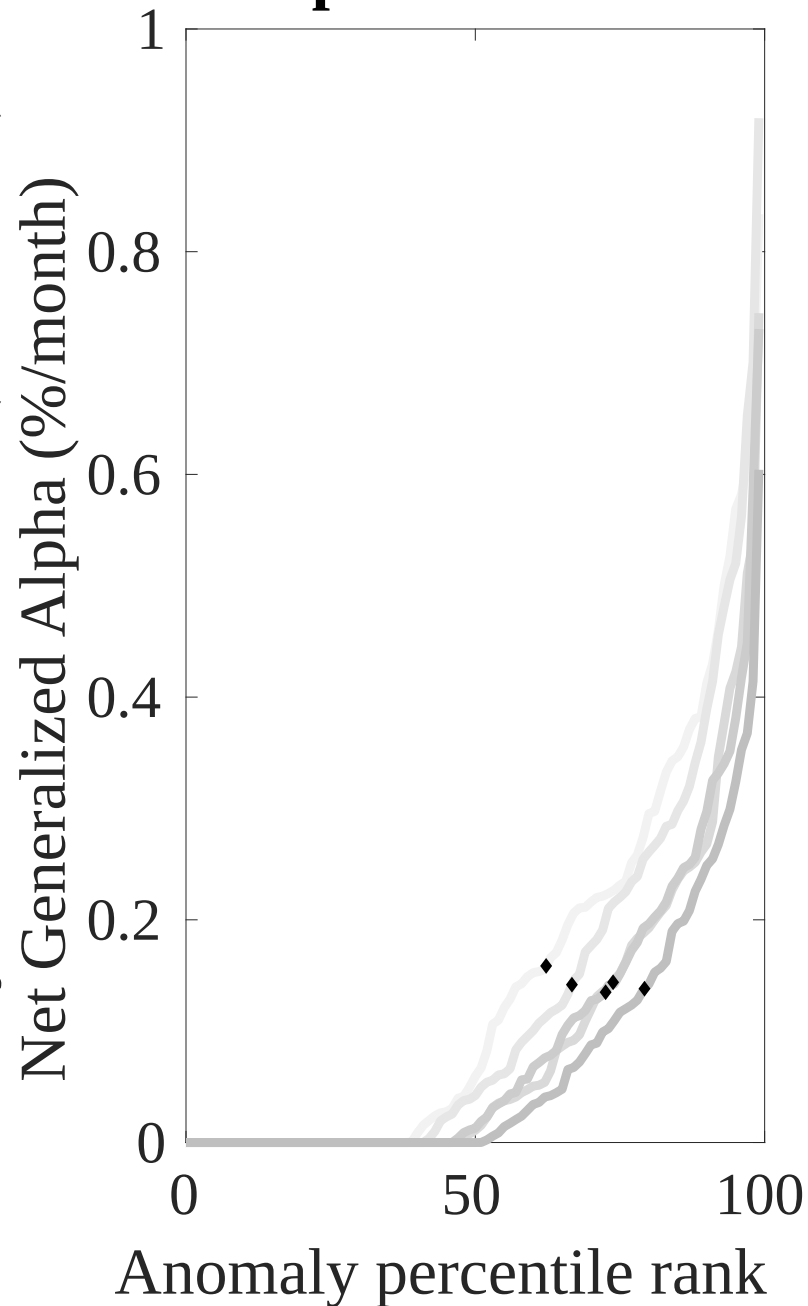
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NIS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



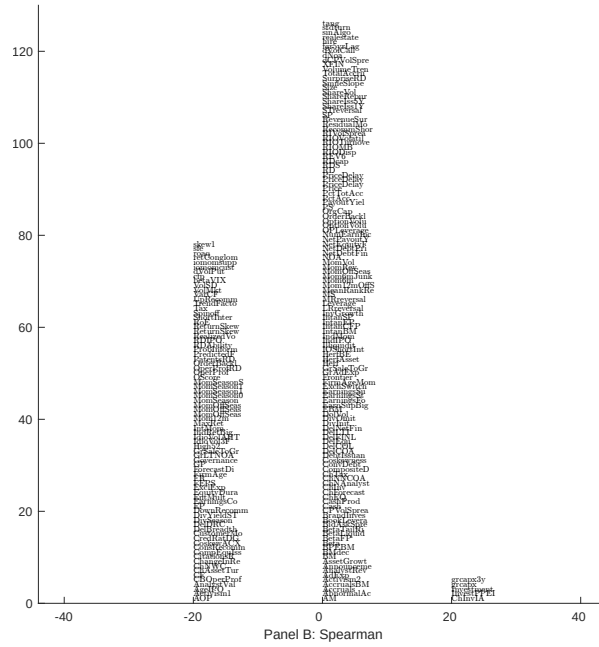
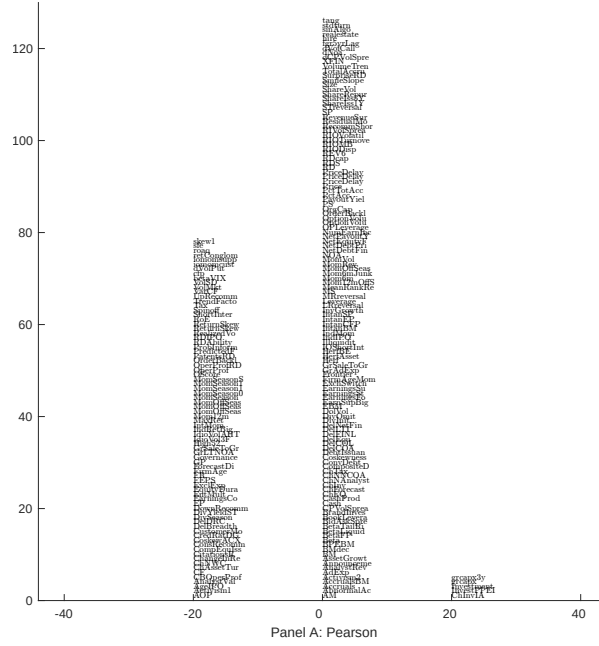


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with NIS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

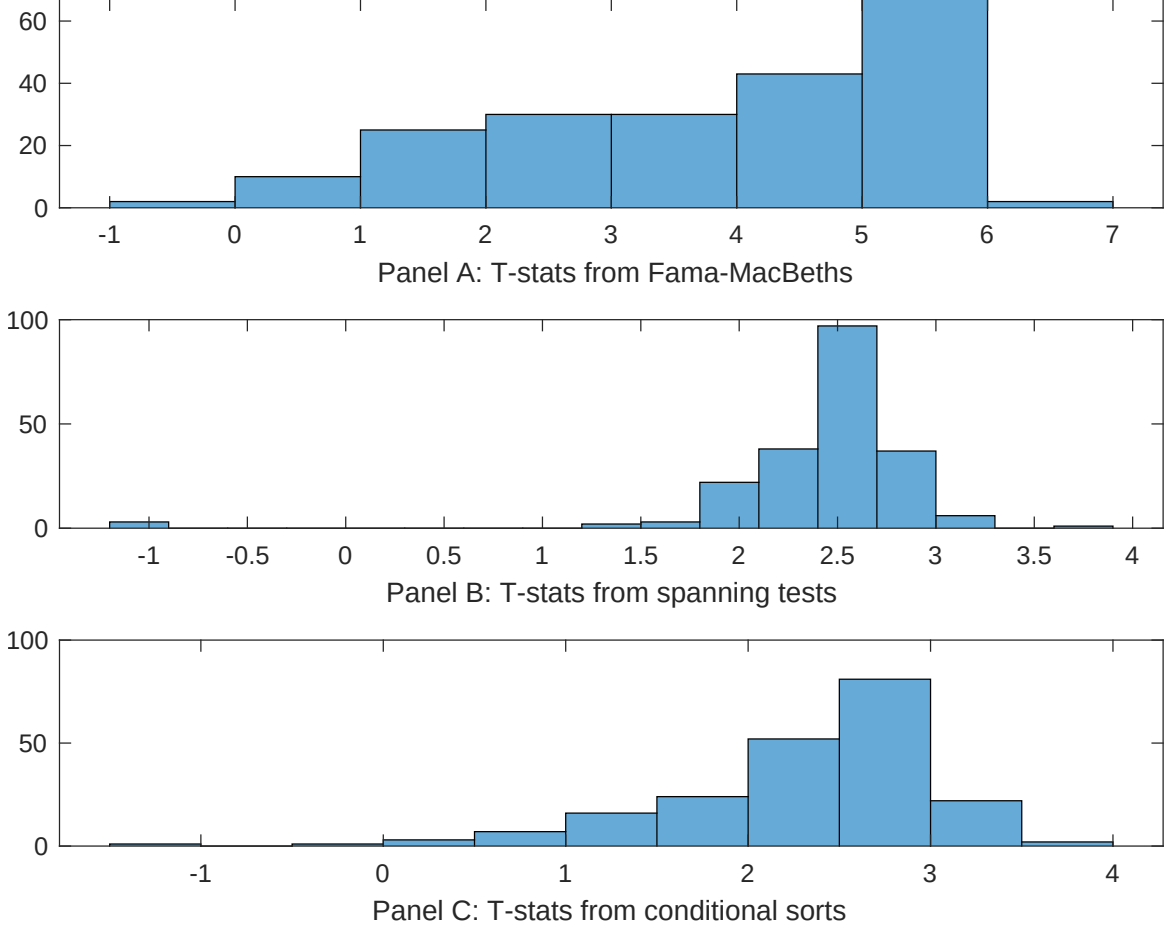


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NIS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.13 [6.07]	0.13 [5.85]	0.17 [6.96]	0.12 [5.47]	0.12 [5.57]	0.11 [5.04]
NIS	0.79 [2.80]	0.96 [3.37]	0.11 [3.26]	0.15 [4.70]	0.16 [5.37]	0.16 [4.77]	0.95 [2.88]
Anomaly 1	0.16 [8.09]						0.41 [1.70]
Anomaly 2		0.98 [8.49]					0.38 [1.78]
Anomaly 3			0.14 [9.45]				0.59 [2.39]
Anomaly 4				0.44 [3.94]			-0.23 [-1.42]
Anomaly 5					0.62 [3.04]		-0.88 [-0.33]
Anomaly 6						0.15 [4.27]	0.77 [1.27]
# months	732	732	720	720	732	720	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NIS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NIS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.88]	0.20 [2.89]	0.18 [2.59]	0.19 [2.76]	0.18 [2.57]	0.19 [2.82]	0.19 [2.73]
Anomaly 1	12.92 [4.01]						10.44 [3.06]
Anomaly 2		15.88 [3.87]					15.35 [3.21]
Anomaly 3			11.08 [2.85]				-2.69 [-0.60]
Anomaly 4				6.00 [1.59]			-0.77 [-0.15]
Anomaly 5					15.30 [4.85]		15.88 [4.29]
Anomaly 6						6.54 [1.90]	-8.32 [-1.57]
mkt	-1.65 [-1.01]	-1.48 [-0.90]	-1.44 [-0.87]	-1.30 [-0.78]	-2.02 [-1.24]	-1.61 [-0.97]	-2.04 [-1.25]
smb	-5.06 [-2.13]	-5.82 [-2.43]	-4.42 [-1.85]	-4.74 [-1.98]	-4.48 [-1.90]	-4.62 [-1.93]	-6.03 [-2.54]
hml	-1.94 [-0.61]	-1.08 [-0.34]	-1.40 [-0.44]	-1.25 [-0.39]	-0.42 [-0.13]	-1.39 [-0.43]	-0.70 [-0.22]
rmw	-5.51 [-1.72]	-5.67 [-1.77]	-5.35 [-1.66]	-5.15 [-1.59]	-3.02 [-0.94]	-4.84 [-1.49]	-4.03 [-1.26]
cma	2.93 [0.55]	-6.33 [-0.93]	4.58 [0.83]	7.03 [1.18]	5.16 [1.05]	6.26 [1.07]	-10.55 [-1.49]
umd	-0.75 [-0.46]	-0.24 [-0.15]	-0.96 [-0.58]	-0.83 [-0.50]	-0.37 [-0.23]	-0.57 [-0.35]	-0.04 [-0.02]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	5	5	4	3	6	3	8

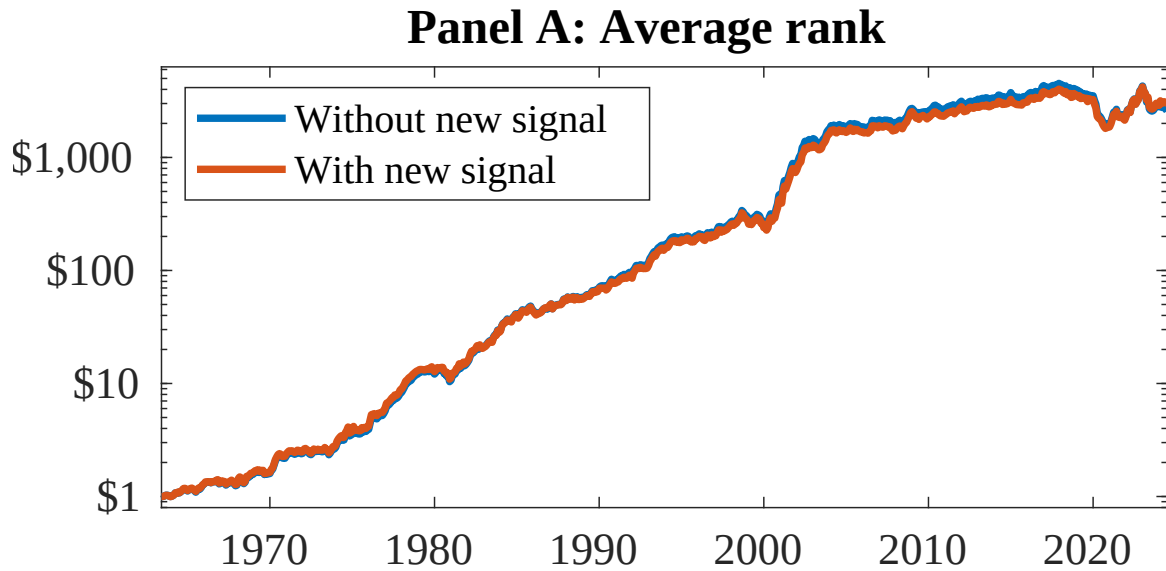


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as NIS. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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